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## ORIGINAL ARTICLE

### Effect of mucoprotective plant-derived therapies on damage to colonic mucosa caused by carprofen and robenacoxib administered to healthy dogs for 21 days

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**Background:** Non-steroidal anti-inflammatory drugs (NSAIDs) may cause gastrointestinal damage in dogs.

**Hypothesis/objectives:** To determine the extent to which lansoprazole, liquorice extract, and a herbal solution exhibit protective effects on colonic mucosa when administered to dogs concurrently with the NSAIDs carprofen or robenacoxib.

**Animals and methods:** Thirty-five healthy beagle dogs (15 male and 20 female) aged 13–14 weeks and weighing 4.3–5.5 kg at the beginning of the experiment were included. Endoscopy and biopsy of the caudal gastrointestinal tract were performed pretreatment and on the last day of a 21-day treatment period with (1) oral carprofen; (2) carprofen and the proton-pump inhibitor lansoprazole; (3) carprofen, liquorice extract, and a herbal solution that contained extracts of thyme, icelandic lichen, hyssop, and saponariae root; (4) robenacoxib; (5) robenacoxib and lansoprazole; (6) robenacoxib, liquorice extract, and herbal solution; or (7) an empty gelatin capsule. Statistical analyses were performed with the Kruskal–Wallis, Cochran's  $Q$ , and chi-squared test with  $p < 0.05$  considered significant.

**Results:** Both carprofen and robenacoxib tested damaged the colonic mucosa with most severe microscopic lesions following administration of robenacoxib with lansoprazole. The risk of histopathological lesions in the colon increased most rapidly in robenacoxib with lansoprazole (absolute risk increase –0.85) similar to robenacoxib only (–0.75), whereas the best result was recorded following the plant remedies together with carprofen (–0.15) and the plant remedies together with robenacoxib (–0.2).

**Conclusions and clinical importance:** Concurrent administration of liquorice extract and an herbal solution with robenacoxib was associated with decreased severity of the NSAID-induced mucosal lesions.

**Keywords:** dog; canine; carprofen; robenacoxib; lansoprazole; herbal treatment; adverse effect; colon

## 1. Introduction

Non-steroidal anti-inflammatory drugs (NSAIDs) are commonly used in dogs to manage chronic pain due to inflammatory joint disease. They have potent analgesic and anti-inflammatory effects and inhibit spinal nociceptive transmission (Miranda et al. 2005). NSAIDs are currently recommended for the prevention and treatment of inflammation, especially for musculoskeletal disorders (Lascelles et al. 2007). The basic mode of action of NSAIDs lies in the inhibition of cyclooxygenase (COX), an enzyme involved in prostaglandin generation (Sakamoto et al. 2006; Lascelles et al. 2007). Prostaglandins are also produced by the gastrointestinal (GI) mucosa and appear to mediate many of the components of what has been termed 'mucosal defense' (Almansori et al. 2005). COX enzymes occur in at least three isoforms: COX-1, COX-2, and COX-3 (Muñoz et al. 2010). Prostaglandins that mediate inflammation are generated by COX-2, while the prostaglandins involved in protecting the GI tract are generated by COX-1 (Bjorkman 2006). COX-1 and COX-2 are both responsible for generating prostaglandins involved with inflammation and those for protection of the GI tract. However, COX-2

prostaglandins are inducible and those prostaglandins produced have more of an inflammatory role (Muñoz et al. 2010). Traditional NSAIDs nonspecifically inhibit both COX-1 and COX-2; their adverse effects on the GI tract are attributed to the inhibition of COX-1–derived prostaglandins. While effective at relieving pain and inflammation, nonselective NSAIDs are associated with a significant risk of serious GI adverse effects (Almansori et al. 2005). Long-term NSAID therapy can generate adverse GI complications ranging from erosions and submucosal hemorrhages to life-threatening complications such as bleeding and perforation (Deneuche et al. 2004). The most common clinical manifestation of NSAID-related colonic mucosal damage in dogs is a combination of erosions and rare ulcerations. However, NSAIDs, along with opioids, have been shown to be efficacious for preventing and treating postoperative pain in dogs (Hunt et al. 2013).

This has prompted a search for safer strategies, including the co-administration of a proton pump inhibitor (PPI) or cotherapy with plant extracts as well as use of prostaglandin E2 analogues (Xie 1996; Blumenthal et al. 2000; Esteve et al. 2005; Paul et al. 2013).

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The recently developed PPI lansoprazole inhibits acid secretion by inhibiting hydrogen/potassium ATPase pump in the apical plasma membrane of parietal cells in the gastric mucosa (Satoh et al. 1989; Arakawa et al. 1990; Nagaya et al. 1990). PPIs are considered to be potent anti-secretory and antiulcer agents on the basis of their comparative efficacy and potency compared to histamine  $H_2$  – receptor antagonists. They are known to have an excellent safety profile (Blandizzi et al. 1990; Masa et al. 2001; Tsuji et al. 2002). In humans, PPIs such as lansoprazole have been shown to be effective in preventing the development of gastric and duodenal ulcers in high-risk patients who are taking NSAIDs. In the clinical experience of the authors, PPI therapy is effective at healing NSAID-induced gastroduodenal ulcers and preventing their recurrence in dogs requiring ongoing NSAID therapy. However, the use of proton pump inhibitors in humans is thought to increase the incidence of microscopic colitis with increased infiltration of intraepithelial lymphocytes, although the exact mechanisms are not fully understood (Yu et al. 2012; Jacques et al. 2013). Still, to our knowledge, the effect of lansoprazole on the distal section of the GI tract has not been analyzed in dogs, which is a rationale for investigating the effect of a PPI on the colon in this study.

Humans have made use of liquorice (mainly *Glycyrrhiza glabra*) as a remedy since the beginning of recorded history. Its use in various geographical regions and time periods has been extensively documented (Armanini et al. 2002). The healing properties of liquorice on the mucosa, which may result from stimulation of mucus synthesis, are well documented (Blumenthal et al. 2000). Liquorice preparations have been studied as a possible adjuvant in treating the spasmodic pains caused by chronic gastritis. They also exhibit anti-inflammatory properties, most likely due to the presence of glycyrrhizinic acid and its metabolite, glycyrrhetic acid generated via deglycyrrhization. The protective actions of a deglycyrrhized liquorice preparation on mucosal damage have been examined previously in dogs (Morris et al. 1974; Langmead & Rampton 2001). The authors in the current study have used liquorice root powder in dogs at an oral dose of 50 mg every 24 hours for dogs up to 10 kg.

The herbal solution used in the present study<sup>1</sup> is a commercially available liquid alcohol extract that consists of thyme, icelandic lichen, hyssop, and saponariae root. The preparation contains many bioactive compounds with potentially therapeutic effects. Icelandic lichen exhibits anti-inflammatory and antibacterial properties and increases the production and secretion of saliva. In appropriate doses, it inhibits gastric acid secretion and prevents the interaction of hydrochloric acid with gastric mucosa. In humans, icelandic lichen is used to treat ulcers, gastric hyperacidity, and inflammation (Langmead & Rampton 2001; Frondoza et al. 2004). After oral administration of the herb hyssop, there is a significant reduction in the number of intestinal bacteria, the binding of bacterial toxins, and a gradual reduction in inflammation. Hyssop reduces smooth muscle tension slightly and restores normal digestive tract peristalsis (Moody et al. 1998;

Poppenga 2002). In humans, the herb thyme is used to prevent and treat peptic ulcers by increasing the secretion of mucus and antibacterial activity (Smart et al. 1986). Saponariae root protectively normalizes the function of the walls of blood vessels and lymphatic vessels in the GI mucosa (Lanas et al. 2009). The aforementioned plant preparations have been used successfully in human medicine to protect the GI mucosa, and the best results are obtained by using extracts of several plants in one preparation (Langmead & Rampton 2001). The authors in the current study have used the herbal solution described above in dogs at an oral dose of 0.1 ml/kg BW every 24 h, diluted 1:5 in warm water.

The present study was designed to investigate the effects of combination treatment with NSAIDs and either a PPI or plant-based preparations with potential gastroprotective properties. The study focuses just at the colon, because there are no studies about large bowel carprofen and robenacoxib toxicity in dogs. We chose carprofen as a poorly selective COX inhibitor and robenacoxib as a selective COX-2 inhibitor. Our objective was to determine the extent to which lansoprazole, liquorice extract, and the herbal solution exhibit protective effects when administered to dogs concurrently with NSAIDs.

## 2. Materials and methods

The study was approved by the Animal Research Local Ethics Committee of the University of Warmia and Mazury, Olsztyn, Poland (number 15/2011 of 30.03.2011). We also received an individual permit (number 1/2011, issued 17 March 2011) for the experiments carried out with animals from the Dean of the Faculty of Veterinary Medicine, University of Warmia and Mazury. All experiments were performed in accordance with the recommendations of the International Council for Laboratory Animal Science and the Council of Europe (Brylińska & Kwiatkowska 1996).

### 2.1. Animals

Thirty-five healthy beagle dogs (15 males and 20 females, 13–14 weeks old and weighing 4.3–5.5 kg at the beginning of the experiment) were recruited in the present study. Before the study, a preliminary clinical and laboratory investigation (complete blood count (CBC) and serum biochemical analyses) was performed to ensure that the dogs were clinically healthy. A parasitological stool examination was also performed. The dogs were dewormed with oxybendazole and niclosamide,<sup>2</sup> febantel, pyrantel pamoate, and praziquantel,<sup>3</sup> and toltrazuril<sup>4</sup> and vaccinated against canine adenovirus type 2, canine parvovirus, and canine distemper.<sup>5</sup> Health checks were made daily during the experiment. The dogs were placed in rooms prepared especially for experimental animals and maintained at room temperature (17–21 °C). They were kept in seven boxes, with five puppies in each box. The rooms were air-conditioned and the boxes were tiled. The cages were equipped with toys and a bed. All dogs were handled and familiarized with the procedure prior to the

study. During the day, the dogs used two runs, which were fenced, covered with a roof, and tiled. All rooms and runs were provided with a water drainage system; twice a day, they were cleaned with high-pressure water containing a 2% solution of disinfectant.<sup>6</sup> The animals were fed with a balanced dog food,<sup>7</sup> which constituted a complete diet for puppies. Water was given *ad libitum*. No mortality occurred in the study and, apart from occasional diarrhea and blood in the feces, no relevant adverse effects or abnormalities were recorded during clinical examinations in dogs from any of the groups.

## 2.2. Experimental protocol and drug treatment

The normal health status of all dogs was confirmed by clinical examination and by hematological and biochemical analyses during an initial acclimatization period of 30 days. The study comprised a three-week treatment. Animals received the first dose five days after the first endoscopy. All treatments were carried out for 21 days and were administered at 1:00–2:00 pm by the same investigator. This investigator performed a general daily health examination for any evidence of vomiting, diarrhea, signs of depression, inappetence, or signs of abdominal pain during the study. The dogs were randomly assigned to one of seven treatment groups ( $n = 5$  in each group): Group 1: 2 mg/kg BW carprofen<sup>8</sup> orally (PO) every 24 h (q 24 h). Group 2: 2 mg/kg BW carprofen, PO, q 24 h and 0.5 mg/kg BW lansoprazole,<sup>9</sup> PO, q 24 h. Group 3: 2 mg/kg BW carprofen, PO, q 24 h; 50 mg/dog liquorice extract, PO, q 24 h<sup>10</sup>; and 0.1 ml/kg BW herbal solution, PO, q 24 h, diluted 1:5 with warm water. Group 4: 2 mg/kg BW robenacoxib,<sup>11</sup> PO, q 24 h. Group 5: 2 mg/kg BW robenacoxib, PO, q 24 h and 0.5 mg/kg BW lansoprazole, PO, q 24 h. Group 6: 2 mg/kg BW robenacoxib, PO, q 24 h; 50 mg/dog liquorice extract, PO, q 24 h; and 0.1 ml/kg BW herbal solution, PO, q 24 h, diluted 1:5 with warm water. The dogs in control Group 7 received an empty gelatin capsule.

The recommended oral dosage of carprofen<sup>8</sup> for dogs is 2–4 mg/kg BW/day. We selected a dose of 2 mg/kg BW according to the information given in the Rimadyl summary of product characteristics for long-term use. Moreover in clinical practice, the authors administered long-term use Rimadyl as a 4 mg/kg BW/day oral dose. Robenacoxib<sup>11</sup> dose for dogs is 1 to 2 mg/kg BW once daily orally. Therapy was administered with robenacoxib at 2 mg/kg BW/day. Based on clinical experience, which indicates that robenacoxib for long-term treatment is used in higher dose than permissible.

## 2.3. Anesthesia

Before the study, blood was collected for a CBC and serum biochemical analyses. The dogs were then weighed and anesthetized for endoscopic evaluation. Food, but not water, was withheld 24 h before anesthesia. In all cases, the anesthetic was administered with a similar protocol for premedication with 0.05 mg/kg BW atropine sulfate<sup>12</sup> and 0.4 mg/kg BW acepromazine maleate.<sup>13</sup> A 24-gauge

catheter was inserted in a cephalic vein. The induction of general anesthesia was performed with a combination of 1 mg/kg BW xylazine<sup>14</sup> and 4 mg/kg BW ketamine,<sup>15</sup> given in the same syringe via the cephalic vein. After the abolition of laryngeal reflexes, all dogs were intubated using a laryngoscope.<sup>16</sup> Maintenance of anesthesia was achieved with isoflurane<sup>17</sup> at a concentration of 1.8%–2%. A 0.9% sodium chloride solution was administered intravenously at 5 ml/kg BW/h throughout the endoscopic procedure. Pain therapy continued for 24 h after surgery with the use of 2–3 mg/kg BW tramadol hydrochloride<sup>18</sup> administered intramuscularly in the hind limb.

## 2.4. Endoscopy and lesion scoring

During the endoscopy, the entire lower bowel up to the caecum was visualized. Colonoscopic examination of the colonic mucosa was done in each group before and on day 21 to ensure mucosal integrity. Teamwork by the hospital personnel and authors ensured that all colonoscopies were performed at both scheduled time-points.

Food was withheld for 24 h and water was withheld for 6 h before anesthesia. To perform the endoscopy, animals were placed in a left lateral recumbent position and a gastrofiberscope<sup>19</sup> (1030 mm probe length, 9.8 mm outer diameter of flange, and 2.8 mm channel diameter) was used. Lesions were graded according to Forsyth: grade 0, no punctate erosions and/or hemorrhaging lesions; grade 1, 1–5 punctate erosions and/or hemorrhaging lesions; grade 2, 6–15 punctate erosions and/or hemorrhaging lesions; grade 3, 16–25 punctate erosions and/or hemorrhaging lesions; grade 4, >25 punctate erosions and/or hemorrhaging lesions and 1–5 invasive erosions; grade 5, >6 invasive erosions; and grade 6, ulcers (Forsyth et al. 1996; 1998). All dogs were grade 0 before treatment. Erosion was defined as a <3 mm diameter discontinuation of the mucosa, and an ulcer was defined as >3 mm with a craterous center. The veterinarian who performed the endoscopies was unaware of the treatments received by the dogs.

## 2.5. Histopathology and lesion scoring

Following the examination of the entire bowel, the endoscope was withdrawn slowly and biopsy samples collected. In areas with no gross lesions, two samples were collected from the ascending, transverse and descending colon. During the second colonoscopy, where specific lesions were detected, samples were collected from the lesion and the surrounding tissue.

Microscopic lesions in the colon were assessed with the following scale: 0, no deviations from healthy tissue; 1, minor lesions; 2, moderate lesions; and 3, severe lesions (0, no obvious changes compared to normal tissue; 1, changes to 30% of the tissue; 2, changes to 31%–60% of the tissue; 3, the most advanced lesions, present in 61%–100% of tissue). Four types of lesions were evaluated according to this scale: infiltration with lymphoid cells to the lamina propria, increased diameter of glands,

proliferation of connective tissue, and hyperemia. The pathologist who examined the samples histologically was blinded to the treatment group.

## 2.6. Preparation of histopathological specimens

The samples were fixed in 10% buffered formalin and were then prepared via the conventional method into paraffin blocks. The microtome sections (4  $\mu$ m) were stained with hematoxylin and eosin (Jacobs et al. 1990; Kleinschmidt et al. 2006).

## 2.7. Statistical analysis

Comparisons of the lesions in the colonic mucosa among the groups were made with the nonparametric Kruskal–Wallis test. Analyses of histopathological lesions in the colonic mucosa were performed with Cochran's  $Q$ -test and chi-squared test and by calculating the  $z$  scores (as the number-independent test). This method allows a comparison between each group in terms of occurrence and the severity of individual histopathological changes. Standardized residues are adjusted to a normal distribution with a mean 0 and standard deviation 1, values greater than 2 are statistically significant. The analysis was performed with Statistica 9.1 (2010 STATISTICA data analysis software system, version 9.1; StatSoft, Inc., www.statsoft.com) under a license issued to the University of Warmia and Mazury in Olsztyn, Poland. A  $p$ -value  $< 0.05$  was considered to be significant.

## 3. Results

### 3.1. Endoscopic examination

The results of endoscopy of the colon on day 0 did not reveal any lesions in the colonic mucosa. The subsequent endoscopic examination, carried out on day 21, showed reddening of the mucosa, as well as erosions and ecchymoses in dogs from I to V treatment groups. Comparisons of colonoscopic findings among the groups did not indicate any statistically significant differences ( $p = 0.2902$ ). However, Group 6 differed least from the control group (Group 7), although this result was not statistically significant. Lesions were most frequently observed in dogs administered carprofen as a single treatment (Group 1), this result was also not statistically significant (Figure 1).

### 3.2. Microscopic examination of the colonic mucosa

On day 0 of the experiment, only single lymphoid cells were observed in most of the examined specimens of colonic mucosa. However, on day 21 of the experiment, the most common lesion was infiltration with inflammatory cells (61% of dogs). Circulatory disturbances were the second most common type of morphological lesion. In 15 dogs, an increase in the diameter of glands was reported; the gland lumens were filled with a substantial amount of mucus and desquamated epithelial cells. All of these lesions were graded as mild. The least common pathological lesion was proliferation of the connective tissue arranged in thick bands situated between the crypts.

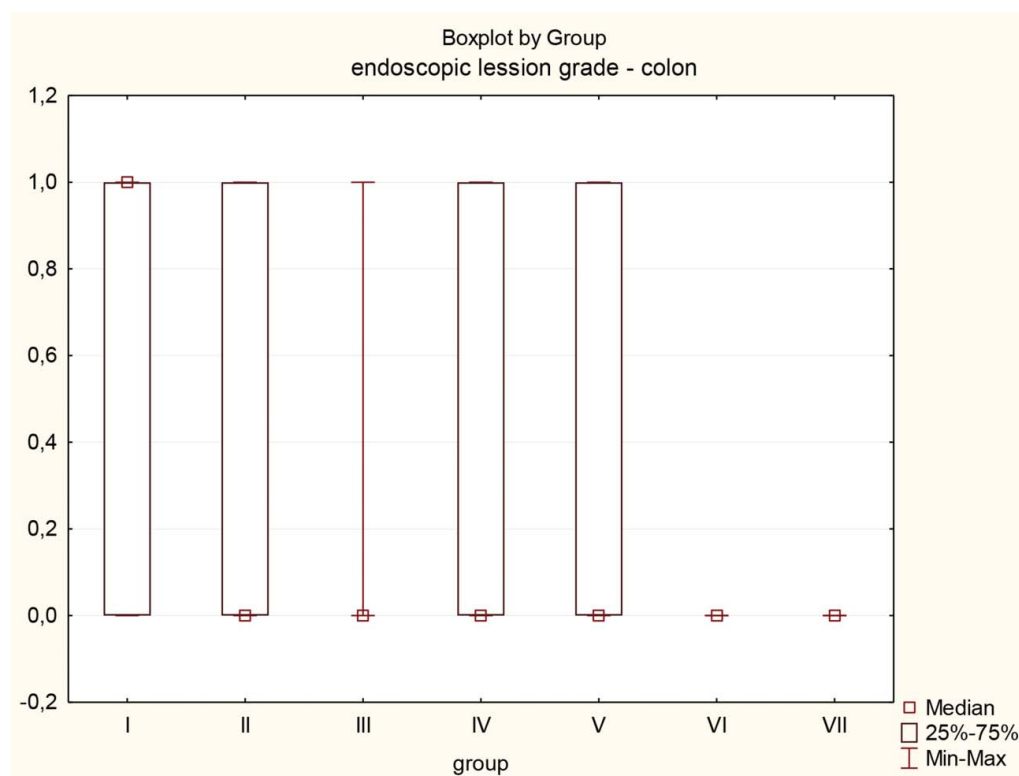


Figure 1. Results of endoscopic examination of the colonic mucosa of dogs in Groups I–VII (I – carprofen, II – carprofen and lansoprazole, III – carprofen, liquorice extract, and herbal solution, IV – robenacoxib, V – robenacoxib and lansoprazole, VI – robenacoxib, liquorice extract, and herbal solution, VII – control group)  $p = 0.2902$ , not statistically significant.



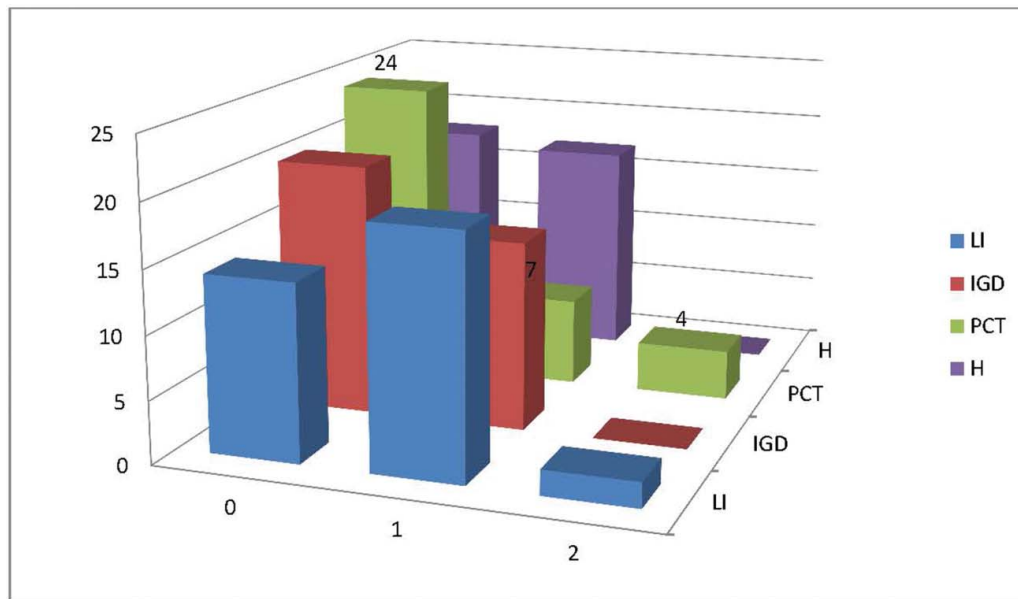


Figure 2. Distribution of histopathological changes in the colonic mucosa at day 21, due to the nature and degree of severity in all dogs together. The scale of assessment of changes: 0 – no deviations from normal tissue, 1 – minor variations, 2 – moderate changes (LI – infiltration with lymphoid cells; IGD – increased diameter of glands; PCT – proliferation of connective tissue; H – hyperemia).

This lesion was confirmed in 31.4% of the examined biopsy specimens (Figure 2).

### 3.3. Analysis of the results of histopathological examination of the colonic mucosa

Cochran's  $Q$ -test yielded statistically significant differences among the groups ( $p < 0.001$ ). The highest number of differences occurred in Group 5 (85%), whereas Group 3 varied the least from the control group (15% difference) (Table 1).

The chi-squared test revealed statistically significant differences among the groups for the variable defined as infiltration of lymphoid cells into the colonic lamina propria. For this parameter, the least favorable results were reported in Group 5, in which the  $z$  scores were statistically significant (2.22). Mild infiltration with lymphoid cells was found in all individuals from Group 5 (Figure 3). This type of lesion was also detected in the biopsy

specimens collected from the majority of dogs administered robenacoxib and carprofen. The results that most closely resembled the findings in the control group occurred in biopsy specimens from dogs that received carprofen together with the plant remedies (Group 3). Mild infiltration with lymphoid cells in the colonic lamina propria was observed in only one individual in this group.

The chi-squared test also revealed statistically significant differences among the groups for the variable defined as an increase in the diameter of glands in the colonic mucosa ( $p = 0.00364$ ). The majority of individuals in which the examined lesion was found were in the groups administered robenacoxib (Group 4) and robenacoxib with lansoprazole (Group 5). An increase in the diameter of glands occurred in only one dog in the group that received carprofen and the plant remedies (Group 3). In all cases, the lesions were mild. This type of lesion was not observed in biopsy specimens from dogs administered robenacoxib and the plant remedies (Group 6). The  $z$  score for this finding was 2.09, which is statistically significant.

The chi-squared test revealed a lack of statistically significant differences among the groups for the variable defined as proliferation of the connective tissue in the colonic mucosa ( $p = 0.10319$ ). Mild proliferation of the connective tissue was found in colonic biopsy specimens from two dogs administered carprofen, two dogs administered carprofen and lansoprazole, and two dogs administered robenacoxib and lansoprazole. The worst result was reported for the biopsy specimens collected from dogs administered robenacoxib alone for 21 d ( $z$  score 2.17, which is statistically significant), whereas the best result was recorded for dogs administered carprofen and the plant remedies ( $z$  score 1.65, statistically significant).

Significant differences among the groups ( $p = 0.00275$ ) were reported for the variable defined as hyperemia of the colonic mucosa. When we took the

Table 1. Scale of microscopic lesions on day 21 divided into groups.

| Group | LI |   |   | IGD |   | PCT |   |   | H |   |
|-------|----|---|---|-----|---|-----|---|---|---|---|
|       | 0  | 1 | 2 | 0   | 1 | 0   | 1 | 2 | 0 | 1 |
| I     | 1  | 4 | 0 | 2   | 3 | 3   | 2 | 0 | 2 | 3 |
| II    | 1  | 3 | 1 | 2   | 3 | 3   | 2 | 0 | 2 | 3 |
| III   | 4  | 1 | 0 | 4   | 1 | 5   | 0 | 0 | 4 | 1 |
| IV    | 0  | 4 | 1 | 1   | 4 | 2   | 1 | 2 | 1 | 4 |
| V     | 0  | 5 | 0 | 1   | 4 | 2   | 2 | 1 | 0 | 5 |
| VI    | 3  | 2 | 0 | 5   | 0 | 4   | 0 | 1 | 4 | 1 |
| VII   | 5  | 0 | 0 | 5   | 0 | 5   | 0 | 0 | 5 | 0 |

Note: LI – infiltration with lymphoid cells; IGD – increased diameter of glands; PCT – proliferation of connective tissue; and H – hyperemia.

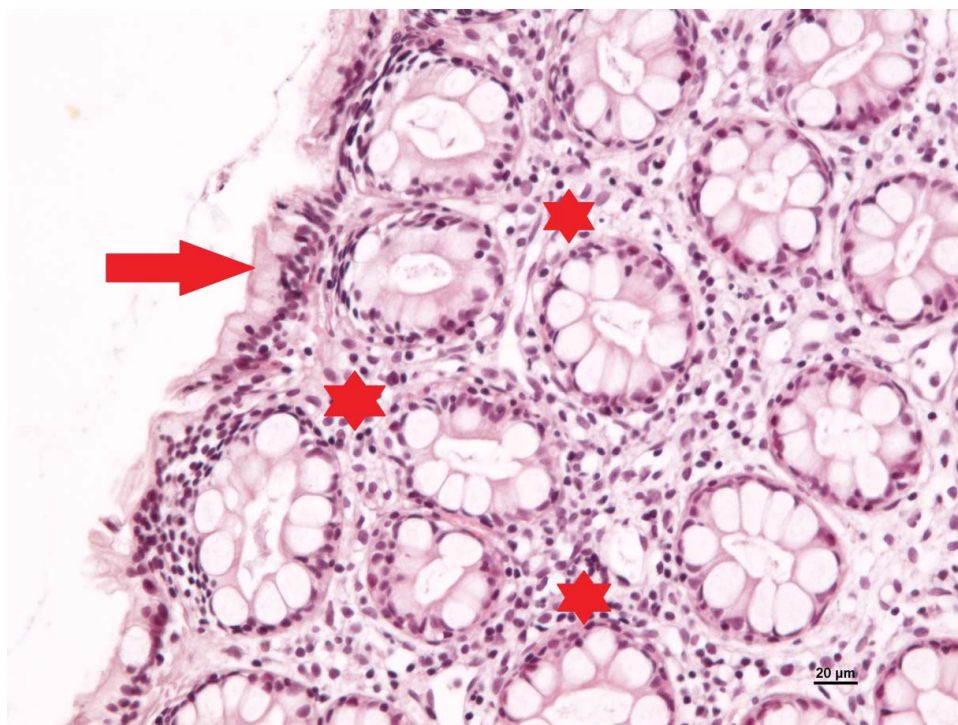


Figure 3. Cross-section of colonic mucosa from a dog in Group 5 on day 21 of the experiment. Diffuse infiltration of inflammatory cells into the lamina propria of the top part of the colonic mucosa (star), and visible exfoliation of mucus at the surface epithelium of the mucosa (arrow). Hematoxylin and eosin staining (magnification  $\times 400$ ).

distribution of the lesions into consideration, we found that most of this variation was due to Groups 4 and 5 (the highest  $z$  score was for Group 5: 2.49). When these two groups were excluded from the analysis, the  $p$ -value was 0.09626; this confirmed that Groups 4 and 5 were responsible for the occurrence of variability, and that they differed most from the control group. This finding was further confirmed by the percentage of hyperemia prevalence in these groups. In Group 4, this type of lesion was found in most of the individuals, whereas in Group 5 it was detected in all biopsy specimens.

According to the percentage distribution of lesions in the groups that were simultaneously administered NSAIDs and plant products (Groups 3 and 6), the plant remedies decreased the risk of hyperemia in the colonic mucosa. In each case of hyperemia, the lesions were mild.

The risk of histopathological lesions in the colon increased most rapidly in Group 5 (absolute risk increase [ARI] =  $-0.85$ ) and was comparable to the risk in Group 4 ( $-0.75$ ), whereas the best result was recorded in the group that was administered the plant remedies together with carprofen (Group 3; ARI =  $-0.15$ ) and with robenacoxib (Group 6; ARI =  $-0.2$ ). In both cases, the confidence interval included the integer 1.

#### 4. Discussion

To the best of our knowledge, this study was the first to compare the protective effects of lansoprazole and plant remedies on colonic mucosa in dogs. Although the effects of concurrent administration of NSAIDs and the plant remedies described herein have not been reported

previously, this regimen has been used by us in clinical practice for dogs taking NSAIDs; no clinical side effects have been noted (Szweda, unpublished observations). Our study, performed under laboratory conditions, also focused on side effects in the distal digestive tract of dogs that received carprofen vs. robenacoxib for long periods of time.

To date, most studies in humans of the adverse effects associated with administration of NSAIDs have been limited to the upper section of the GI tract; this has resulted in the belief that the most severe (even lethal) damage affects the stomach and the duodenum. However, larger scale studies in humans have indicated that NSAIDs may also increase the risk of damage to the lower section of the GI tract (Bjarnason et al. 1993; Fujimori et al. 2008, 2009). Singh and Triadafilopoulos (2005) reported that the introduction of endoscopy to human gastroenterology has enabled clinicians to observe that lesions (bleeding, erosions, and perforations) also develop in the large bowel during treatment with NSAIDs. The results of our study provide further confirmation of these observations.

Endoscopy is considered a sensitive method for detecting early NSAID-induced GI injuries. Endoscopy may identify GI lesions in dogs, even if they do not present any clinical signs of side effects due to NSAID use (Luna et al. 2007). None of the dogs in the present study developed serious endoscopic lesions, but the absence of side effects could have been due to the small number of animals used in this study. This information cannot be extrapolated to a clinical setting when often patients may have other concurrent diseases. In such circumstances, GI lesions may be exacerbated and result in clinical consequences. Advanced endoscopic modalities that allow for

full investigation of the entire GI tract have revealed that NSAIDs can cause a variety of abnormalities (Masa et al. 2001; Goldstein et al. 2005). Several reports have revealed that NSAIDs can induce colonic injuries in dogs (Jackson et al. 2000; Lans et al. 2007). As expected, the preventive effect of PPIs did not extend to the colon in our study. The endoscopic lesions in the GI tract of the young beagles in the present study were evidence that both of the NSAIDs that we tested caused damage to the colonic mucosa. Numerous endoscopic studies in humans have indicated that preferential NSAIDs can damage the mucosa of the lower GI tract. Hence, it was not surprising that we found lesions in the colons of dogs that were administered robenacoxib both alone and in combination with lansoprazole. The most severe damage to the colonic mucosa was observed during endoscopic examination of dogs that were administered carprofen, which is associated with inhibition of both COX-1 and COX-2 (Jackson et al. 2000; Lans et al. 2007).

In a previous study in young dogs aged 5–6 months, the administration of significantly higher doses of robenacoxib than those used in the present study was not associated with microscopic lesions in the colonic mucosa (King et al. 2010). Moreover, Blois et al. (2010) reported that the use of carprofen (4.4 mg/kg BW, PO, q 24 h) in adult dogs for four weeks did not cause significant lesions in the GI mucosa. These results are inconsistent with those of our study.

Based on our findings and those of some other researchers, there appears to be a need for medicinal compounds with protective actions on the GI tract of dogs. The protective effects on the colonic mucosa of the herbal solution used in the present study have been confirmed in studies with humans (Chen & Zhang 1990; Moody et al. 1998). There have been few studies of the effects of this herbal solution in dogs, but several studies have indicated that the specific plant extracts contained in the herbal solution are successfully used in veterinary medicine for the treatment of GI disorders (Blumenthal et al. 2000; Poppenga 2002; Viegli et al. 2003; Lanan et al. 2009). In the present study, we found that only dogs that received the herbal solution, liquorice extract together with robenacoxib showed no signs of damage to the colonic mucosa. Liquorice has been used as an alternative to PPIs in the treatment of gastric ulcers in human medicine, as well as to treat large bowel ulcerations (Bernardi et al. 1994; Wang et al. 2000). Morris et al. (1974) found that liquorice has protective properties with regard to the gastric mucosa. The results yielded by our study support those of Morris et al.

Our results suggest that the use of the two plant remedies included in our study may, when administered concurrently with NSAIDs, reduce the risk of NSAID-induced hyperemia in the colonic mucosa of young beagle dogs.

## 5. Conclusions

In the present study, we demonstrated that carprofen and robenacoxib induced lesions in the colons of young,

healthy beagles when administered for 21 days. However, when robenacoxib was administered concurrently with an herbal solution and liquorice extract, no damage to the colonic mucosa was observed by endoscopy. This combination was also associated with the least severe histopathological lesions in colonic biopsy specimens. The most extensive microscopic colonic lesions were associated with robenacoxib administered together with lansoprazole. The plant remedies that we tested appeared to reduce the severity of NSAID-induced damage to the colonic mucosa; this was observed during both colonoscopy and histopathological examination. As a consequence, we use herbal remedies in combination therapy with NSAIDs in dogs. Our results do not necessarily apply to all dogs because it involved only young, healthy beagle dogs. Further studies in dogs to measure the pharmacokinetics of the herbal solution and liquorice extract throughout the distribution and elimination phases are needed to confirm the suitability of this protocol for prolonged use in dogs. Because the low numbers of dogs in each treatment group further studies are needed to confirm the suitability of the proposed protocol used in this experiment.

## Notes

1. Pectosol; Herbol Pruszków, Pruszków, Poland.
2. Vetminth; Virbac, Parice, France.
3. Drontal Plus flavor; Bayer, Luverkusen, Germany.
4. 5% Baycox; Bayer, Luverkusen, Germany.
5. NOBIVAC Puppy; NOBIVAC DHP, Boxmeer, Netherlands.
6. Virkon S Du Pont Company, Antec International Limited, Suffolk, United Kingdom.
7. Puppy Junior Diet with Fish; FORZA 10, Bagnoli di Sopra, Italy.
8. Rimadyl; Pfizer, Warsaw, Poland.
9. LansoLek; Sandoz GmbH, Kundl, Austria.
10. Lukrecja; Herbol Pruszków, Krakow, Poland.
11. Onsiar; Novartis, Surrey, United Kingdom.
12. Atropinum sulfuricum; Polfa, Warsaw, Poland.
13. Calmivet; Vétérinaire, Lure, France.
14. Rometar; SPOFA, Prague, Czech Republic.
15. Bioketan; Biovet, Gorzów Wielkopolski, Poland.
16. Heine; Herrsching, Germany.
17. Isoflurane; Abbott, Maidenhead, Great Britain.
18. Tramal 50; Polpharma SA, Starogard Gdański, Poland.
19. Olympus GIF XQ-30; Olympus, Philadelphia, PA, USA.

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